NESTED LOCALISING TENSOR IDEALS GIVE ISOMORPHIC WEAK RINGS OF THE ASSOCIATED PURE-INJECTIVES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/jep-2020-135-homological-support/full-original-native.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the retained preamble is the source's own retained-preamble.tex, which carries the author's macro and theorem declaration block verbatim (its own header records which lines of the pinned file it copies); the wrapper class supplies the baseline packages that block assumes. Nothing was added, no macro was invented, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. The displayed numbers of those pointers are the author's own numbering under the author's \newtheorem declarations (the source has no subsections, so numbering is section.index), computed mechanically by the lane's scanner.

Editorial note. Four references to other labelled items occur inside the counted spans (the construction Cons:E_C and the propositions Prop:ring-in-AB, Prop:A/C and its clause it:A/C-d); each is recorded as a navigation stub with a kind from the assembler's enumeration and the author's own number. The labels containing a slash are exactly the ones that the assembler's whitelist now admits.

Editorial note. The counted spans contain no bibliography citation, so the bibliography staged with the source is attached unchanged and no per-card excerpt was needed.

Editorial note. Assembly history: the first attempt (r001) failed with 'Environment Prop undefined' because the preamble carried only this source's retained-preamble.tex, which for this source is a bare layout adapter and contains no ewtheorem declarations; r002 adds the author's own macro and theorem declaration block (lines 6-143 of the pinned file). The rule was refined accordingly: a retained-preamble only replaces the author block when it actually contains theorem declarations. No statement, span or stub changed between the revisions.

Allowed standard prerequisites (editorial).

- The weak ring attached to a localising tensor ideal. Same source, Construction labelled Cons:E_C; invoked in the counted statement at line 507. Supplies the objects EB and EC whose isomorphism is asserted.
- Units of the weak rings in the intermediate quotient are monomorphisms. Same source, Proposition labelled Prop:ring-in-AB; invoked inside the counted proof at line 524. Identifies the unit of the intermediate weak ring with a monomorphism, which lets the injective hull of the unit be compared.

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- Isomorphisms in the intermediate quotient that descend to the big category. Same source, Proposition labelled Prop:A/C, clause (it:A/C-d); invoked at the end of the counted proof at line 533. Lifts the isomorphism produced in the intermediate quotient back to the ambient tensor-triangulated category and makes it compatible with the units.

Source reference. Original definition 6: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 507–507. Navigation pointer for \ref{Cons:E_C} used inside the counted span (line 507); the argument that consumes it is retained in the span.

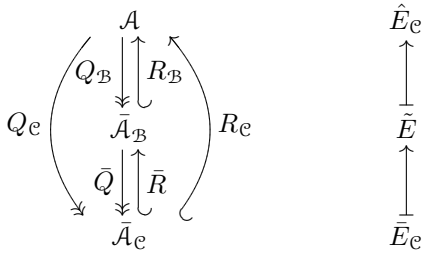
Source reference. Original proposition 4: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 533–533. Navigation pointer for \ref{Prop:A/C} used inside the counted span (line 533); the argument that consumes it is retained in the span.

Source reference. Original proposition 5: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 524–524. Navigation pointer for \ref{Prop:ring-in-AB} used inside the counted span (line 524); the argument that consumes it is retained in the span.

Source reference. Original proposition 4: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 533–533. Navigation pointer for \ref{it:A/C-d} used inside the counted span (line 533); the argument that consumes it is retained in the span.

0.1. **Proposition.** *Let $\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^{\mathrm{c}})$ and $\mathcal{C} \subset \mathcal{A}$ be a proper localizing $\otimes$-ideal, containing $\mathcal{B}$ or equivalently containing $\mathrm{Loc}(\mathcal{B})$. Let $E_{\mathcal{B}}$ and $E_{\mathcal{C}}$ be the corresponding weak rings of Original definition 6. Then there is an isomorphism $E_{\mathcal{B}} \simeq E_{\mathcal{C}}$ in $\mathcal{T}$ compatible with the unit maps $\mathbb{1} \to E_{\mathcal{B}}$ and $\mathbb{1} \to E_{\mathcal{C}}$.*

Proof. Since $\mathrm{Loc}(\mathcal{B}) \subseteq \mathcal{C}$, we can perform the Gabriel quotient $\mathcal{A}/\mathcal{C}$ in two steps:



for an intermediate Gabriel localization $\bar{Q} \dashv \bar{R}$. Let $\tilde{E} := \bar{R}(\bar{E}_{\mathcal{C}})$ the injective object in $\bar{\mathcal{A}}_{\mathcal{B}}$ associated to the injective hull $\bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ in $\bar{\mathcal{A}}_{\mathcal{C}}$. By construction of $E_{\mathcal{C}} \in \mathcal{T}$, we have $\hat{E}_{\mathcal{C}} \cong R_{\mathcal{C}}(\bar{E}_{\mathcal{C}}) \cong R_{\mathcal{B}}(\tilde{E})$ and therefore $\tilde{E} \cong Q_{\mathcal{B}}(\hat{E}_{\mathcal{C}}) = \bar{\mathbb{h}}_{\mathcal{B}}(E_{\mathcal{C}})$. Since $E_{\mathcal{C}}$ is a weak ring in $\mathcal{T}$, its image $\tilde{E}$ is a weak ring in $\bar{\mathcal{A}}_{\mathcal{B}}$, that is non-zero since $\bar{Q}(\tilde{E}) \cong \bar{E}_{\mathcal{C}} \neq 0$. By Original proposition 5, the unit $\bar{\mathbb{h}}_{\mathcal{B}}(\eta_{\mathcal{C}})\colon \bar{\mathbb{1}} \rightarrowtail \tilde{E}$ is therefore a monomorphism in $\bar{\mathcal{A}}_{\mathcal{B}}$. On the other hand, $\bar{\mathbb{h}}_{\mathcal{B}}(\eta_{\mathcal{B}})\colon \bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{B}}$ is by definition the injective hull of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. Hence there exists a commutative diagram in $\bar{\mathcal{A}}_{\mathcal{B}}$

$$\begin{array}{ccc} \bar{\mathbb{1}} & \xrightarrowtail{\bar{\mathbb{h}}_{\mathcal{B}}(\eta_{\mathcal{B}})} & \bar{E}_{\mathcal{B}} \\ & \searrow^{\bar{\mathbb{h}}_{\mathcal{B}}(\eta_{\mathcal{C}})} & \big\downarrow{\scriptstyle \exists\, \varphi} \\ & & \tilde{E} \end{array} \tag{0.2}$$

where φ is a split monomorphism. Applying the exact functor $\bar{Q}$ to this diagram and using that $\bar{Q}(\tilde{E}) = \bar{E}_{\mathcal{C}}$ is the injective hull of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{C}}$, we see that $\bar{Q}(\varphi)$ must

be an isomorphism. In summary, $\tilde{E} \cong \bar{E}_{\mathcal{B}} \oplus N$ in $\bar{\mathcal{A}}_{\mathcal{B}}$ for an object N such that $\bar{Q}(N) = 0$. But since $\tilde{E} = \bar{R}(\bar{E}_{\mathcal{C}})$ and $\bar{R}$ is fully faithful, this forces the object N to be in the image of $\bar{R}$ as well (it is '$\bar{R}$-local'). It follows that $N \cong \bar{R}\bar{Q}(N) = 0$. Therefore φ is already an isomorphism $\bar{E}_{\mathcal{B}} \simeq \tilde{E} \simeq \bar{R}(\bar{E}_{\mathcal{C}})$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. By Original proposition 4 (4) (applied to $\bar{\mathcal{A}} = \mathcal{A}/\operatorname{Loc}(\mathcal{B})$) the isomorphism φ in (0.2) comes from $\mathcal{T}$ and is compatible with the units there. □

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